Bachelor of Information Technology
First Semester
C Programming

Lab Questions

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Pattern:

1. N

   NE

   NEP

   NEPA

   NEPAL

```
#include<stdio.h>
int main()
{
        int i,j;
        char p[6]="NEPAL";
        for (i=0; i<5; i++)
        {
                for (j=0; j<=i; j++)
                {
                        printf("%c",p[j]);
                }
                printf("\n");
        }
        return 0;
}
```

2. 5 4 3 2 1
   5 4 3 2
   5 4 3
   5 4
   5

```
#include<stdio.h>
int main()
{
        int i,j;
        for (i=1; i<=5; i++)
        {
                for (j=5; j>=i; j--)
                {
                        printf("%d",j);
                }
                printf("\n");
        }
        return 0;
}
```

3.

```
        *
      * * *
    * * * * *
  * * * * * * *
* * * * * * * * *
```

```
#include<stdio.h>
int main()
{
        int i,j,k;
        for (i=0; i<=4; i++)
        {
                for (k=3; k>=i; k--)
                {
                        printf("  "); //two spaces
                }
                for (j=0; j<=i*2; j++)
                {
                        printf("* ");
                }
                printf("\n");
        }
        return 0;
}
```

4.

```
* * * * * * * * *
  * * * * * * *
    * * * * *
      * * *
        *
```

```
#include<stdio.h>
int main()
{
	int i,j,k;
	for (i=4; i>=0; i--)
	{
		for (k=4; k>i; k--)
		{
			printf("  "); //two spaces
		}
		for (j=0; j<=i*2; j++)
		{
			printf("* ");
		}
		printf("\n");
	}
	return 0;
}
```

5.

```
$  *  $  *
*  $  *  $
$  *  $  *
*  $  *  $
```

```
#include<stdio.h>
int main()
{
	int i,j;
	for (i=1; i<=4; i++)
	{
		for (j=1; j<=4; j++)
		{
			if ((i+j)%2==0)
			{
				printf("# ");
			}
```

```
                else
                {
                        printf("* ");
                }
        }
        printf("\n");
    }
    return 0;
}
```

String:

6. Write a program to print ASCII values of each and every character of the string given by the user.

```
#include<stdio.h>
int main()
{
        int i;
        char m[50];
        printf("enter your word\n");
        scanf("%s",&m);
        for(i=0; m[i]!='\0'; i++)
        {
                printf("%c gives %d in ASCII\n",m[i],m[i]);
        }
        return 0;
}
```

7. Write a program to reverse a string without using string handling function.

```
#include<stdio.h>
int main()
{
        int i,total=-1;
        char m[50],n[50];
        printf("enter your word\n");
        scanf("%s",&m);
        for(i=0; m[i]!='\0'; i++)
        {
                total++;
        }
        for(i=0; i<=total; i++)
        {
                n[total-i]=m[i];
        }
    n[total+1]='\0';
        printf("reverse of %s is: %s",m,n);
        return 0;
}
```

8. Write a program to compare whether two strings are identical or not without using string handling function.

```
#include<stdio.h>
```

```
int main()
{
	int i,count1=0,count2=0,total=0;
	char m[50],n[50];
	printf("enter your 1st word\n");
	scanf("%s",&m);
	printf("enter your 2nd word\n");
	scanf("%s",&n);
	for(i=0;m[i]!='\0'; i++)
	{
		count1++;
	}
	for(i=0; n[i]!='\0'; i++)
	{
		count2++;
	}
	if (count1==count2)
	{
		for(i=0; m[i]!='\0'; i++)
		{
			if (n[i]==m[i])
			{
				total++;
			}
		}
		if (count1==total)
		{
			printf("the strings are identical\n");
		}
		else
		{
			printf("the strings are not identical\n");
		}
	}
	else
	{
		printf("the strings are not identical\n");
	}
	return 0;
}
```

9. Write a program to count no of spaces, vowels in a sentence input by the user.

```
#include<stdio.h>
int main()
{
	int vcount=0,scount=0,i;
	char s[100];
	printf("enter your sentence:\n");
	gets(s);
	for (i=0;s[i]!='\0';i++)
	{
		if (s[i]=='A'||s[i]=='a'||s[i]=='E'||s[i]=='e'||s[i]=='I'||s[i]=='i'||s[i]=='O'||s[i]=='o'||s[i]=='U'||s[i]=='u')
		{
			vcount++;
		}
		else if (s[i]==' ')
		{
			scount++;
		}
	}
	printf("In the given sentence there are %d vowels and %d spaces\n",vcount,scount);
	return 0;
}
```

10. Write a program to convert a lower-case string to upper case string without using string handling function.

```
#include<stdio.h>
int main()
{
        int i;
        char s[100];
        printf("enter your sentence:\n");
        gets(s);
        for (i=0;s[i]!='\0';i++)
        {
                if (s[i]<='Z'&&s[i]>='A')
                {
                        s[i]=s[i]-32;
          }
        }
        printf("Your new sentence is %s\n",s);
        return 0;
}
```

11. Write a program to combine two different words into one without using string handling function (e.g.: word 1: kist, word 2: college, Result: kist college)

```
#include<stdio.h>

int main()

{

        int i,count1=0,count2=0;

        char s1[50],s2[50];

        printf("enter your first word:\n");

        scanf("%s",&s1);

        printf("enter your second word:\n");

        scanf("%s",&s2);

        for (i=0;s1[i]!='\0';i++)

        {

          count1++;

        }

        for (i=0;s2[i]!='\0';i++)

        {

          count2++;

        }

        char s3[count1+count2+2];

        for (i=0;i<=count1-1;i++)

        {

                s3[i]=s1[i];

        }

        for (i=0;i<=count2-1;i++)
```

```
        {
                s3[count1+1+i]=s2[i];
        }
        s3[count1]=' ';
        s3[count1+count2+1]='\0';
        printf("The merge of %s and %s is %s",s1,s2,s3);
        return 0;
}
```

Array:

12. Write a program to take N numbers from user and only display numbers which are prime.

```
#include<stdio.h>
int main()
{
        int i,j,n,count;
        printf("how many numbers do you want to store?\n");
        scanf("%d",&n);
        int num[n];
        for (i=0;i<n;i++)
        {
                printf("enter number [%d]:",i);
                scanf("%d",&num[i]);
        }
        for (i=0;i<n;i++)
        {
                count=0;
                for (j=1;j<=num[i];j++)
                {
                        if ((num[i]%j)==0)
                        {
                                count++;
                        }
        }
        if (count==2)
        {
                printf("%d\t",num[i]);
                }
        }
        return 0;
}
```

13. Write a program to take N numbers from user and arrange them in reverse order.

```
#include<stdio.h>
int main()
{
        int i,j,n,rev;
        char input;
        printf("how many numbers do you want to store?\n");
        scanf("%d",&n);
        int num[n];
        for (i=0;i<n;i++)
        {
                printf("enter number [%d]:",i);
```

```
                scanf("%d",&num[i]);
        }
        for (i=0;i<n;i++)
        {
                for (j=num[i];j>0;j=j/10)
                {
                  rev=j%10+rev*10;
        }
        printf("rev of %d is %d\n",num[i],rev);
        printf("do you want to store the reversed value?(Y/N)");
        scanf("%c",&input);
        if (input=='Y'||input=='y')
        {
                num[i]=rev;
                }
                rev=0;
        }
        return 0;
}
```

14. Write a program to take N numbers from user and count how many even and odd numbers are present.

```
#include<stdio.h>
int main()
{
        int i,n,even=0,odd=0;
        printf("how many numbers do you want to store?\n");
        scanf("%d",&n);
        int num[n];
        for (i=0; i<n; i++)
        {
                printf("enter number [%d]:",i);
                scanf("%d",&num[i]);
        }
        for (i=0; i<n; i++)
        {
                if (num[i]%2==0)
                {
                        even++;
                }
                else
                {
                        odd++;
                }
        }
        printf("There are %d even\n",even);
        printf("There are %d odd\n",odd);
        return 0;
}
```

15. Write a program to take 3x3 matrix input from user and calculate sum of diagonal elements.

```
#include<stdio.h>
int main()
{
        int i,j,diag=0,num[3][3];
        for (i=0; i<=2; i++)
        {
                for (j=0; j<=2; j++)
                {
                        printf("enter number [%d][%d]:\n",i,j);
                        scanf("%d",&num[i][j]);
                        if (((i+j)%2)==0 || (i==0 && j==0))
                        {
```

```
                    diag=diag+num[i][j];
                }
            }
        }
        printf("The sum of diagonal is:%d\n",diag);
        return 0;
}
```

16. Write a program to take 3x3 matrix input from user and calculate sum of four corners element.

```
#include<stdio.h>
int main()
{
        int i,j,m[3][3],sum=0;
        for (i=0; i<3; i++)
        {
            for  (j=0; j<3; j++)
            {
                printf("enter  element  %d X %d:",i+1,j+1);
                scanf("%d",&m[i][j]);
                if ((i+1)%2!=0&&(j+1)%2!=0)
                {
                    sum=sum+m[i][j];
                }
            }
        }
        printf("sum of four corners of the martix is %d",sum);
        return 0;
}
```

17. Write a program to take two 3X3 matrix, add them and display transpose of final result.

```
#include<stdio.h>
int main()
{
        int i,j,num1[3][3],num2[3][3],transpose[3][3];
        for (i=0; i<=2; i++)
        {
            for (j=0; j<=2; j++)
            {
                printf("enter number [%d][%d] for matrix 1:\n",i,j);
                scanf("%d",&num1[i][j]);
                printf("enter number [%d][%d] for matrix 2:\n",i,j);
                scanf("%d",&num2[i][j]);
                transpose[j][i]=num1[i][j]+num2[i][j];
            }
        }
        printf("The transpose of sum of matrices is:\n");
        for(i=0; i<=2; i++)
        {
            for(j=0; j<=2; j++)
            {
                printf("%d\t",transpose[i][j]);
            }
            printf("\n");
        }
        return 0;
}
```

18. Write a program to take 10 numbers from user and display the numbers in ascending order, counting total number of digits which are odd.

```
#include<stdio.h>
int main()
{
	int i,j,num[10],count=0,temp;
	for (i=0; i<10; i++)
	{
		printf("enter number %d:\n",i+1);
		scanf("%d",&num[i]);
		if((num[i]%2)!=0 && num[i]!=0)
		{
			count++;
		}
	}
	printf("number of odd numbers is:%d\n",count);
	printf("the series is:");
	for(i=0; i<10; i++)
	{
		for(j=i+1; j<10; j++)
		{
			if (num[i]>num[j])
			{
				temp=num[i];
				num[i]=num[j];
				num[j]=temp;
			}
		}
		printf("%d ",num[i]);
	}
	return 0;
}
```

Recursion:

19. Write a program to find sum of Natural Numbers Using Recursion.

```
#include<stdio.h>
int rec(int);
int main()
```

```
{
        int i;
        printf("enter number of iteration for sum of natural numbers:\n");
        scanf("%d",&i);
        printf("sum of natural %d numbers is:%d",i,rec(i));
        return 0;
}
int rec(int j)
{
        int sum;
        if (j!=1)
        {
                return j+ rec(j-1);
        }
}
```

20. Write a program to generate Fibonacci Series using recursion.

```
#include<stdio.h>
void fibo(int);
int n=0,m=1,p=0;
int main()
{
        int i;
        printf("enter number of iteration for fibonacci do you need?\n");
        scanf("%d",&i);
        rec(i);
        return 0;
}
void rec(int j)
{
        if (j!=0)
        {
                printf("%d ",n);
                n=n+m;
                m=p;
                p=n;
                rec(j-1);
        }
}
```

21. Write a program to find the power of a given number using recursion.

```
#include<stdio.h>
int power(int,int);
int main()
{
        int i,j;
        printf("enter exponent of the number:\n");
        scanf("%d",&i);
        printf("enter base of the number:\n");
        scanf("%d",&j);
        printf("power of exponent %d and base %d is:%d",i,j,power(i,j));
        return 0;
}
int power(int e,int b)
{
        if (e==0)
        {
                return 1;
        }
        else
        {
                return b*power(e-1,b);
```

```
        }
}
```

22. Write a program to find factorial of a given number using recursion.

```
#include<stdio.h>
int fact(int);
int main()
{
        int i;
        printf("enter the number to calculate the factorial:\n");
        scanf("%d",&i);
        printf("%d is the factorial of %d",i,fact(i));
}
int fact(int f)
{
        if (f==0)
        {
                return 0;
        }
        else if (f!=1)
        {
                return f*fact(f-1);
        }
        else if (f==1)
        {
                return 1;
        }
}
```

Pointers:

23. Write a program to add all the even numbers in an array and display result using pointers only.

```
#include<stdio.h>
int main()
{
        int n,*p,add=0,*q,i;
        p=&n;
        q=&add;
        printf("how many numbers do you want to store?\n");
```

```
        scanf("%d",p);
        int a[*p];
        for (i=0;i<*p;i++)
        {
                printf("enter your number %d:\n",i+1);
                scanf("%d",a+i);
                if (*(a+i)%2==0)
                {
                        *q=*q+*(a+i);
                }
        }
        printf("sum of even numbers in the array is:%d",*q);
        return 0;
}
```

24. Write a program to find highest value amongst 10 numbers given by user using pointers.

```
#include<stdio.h>
int main()
{
        int great,*g,a[10],i;
        g=&great;
        for (i=0; i<10; i++)
        {
                printf("enter your number %d:\n",i+1);
                scanf("%d",a+i);
                if (i==0)
                {
                        *g=*a;
                }
                if (*(a+i)>*g)
                {
                        *g=*(a+i);
                }
        }
        printf("highest value in the array is:%d",*g);
        return 0;
}
```

25. Write a program to enter 10 numbers and calculate even and odd numbers count using pointers only.

```
#include<stdio.h>
int main()
{
        int even=0,odd=0,*e,*d,a[10],i;
        e=&even;
        d=&odd;
        for (i=0; i<10; i++)
        {
                printf("enter your number %d:\n",i+1);
                scanf("%d",a+i);
                if (*(a+i)%2==0)
                {
                        (*e)++;
                }
                else
                {
                        (*d)++;
                }
        }
        printf("there are %d even number(s) and %d odd number(s) in the array.",*e,*d);
        return 0;
}
```

26. Write a program to add two arrays and count number of odd numbers in the final result using pointers only,

```
#include<stdio.h>
int main()
{
	int even=0,odd=0,*e,*d,a[10],b[10],c[10],i;
	e=&even;
	d=&odd;
	for (i=0; i<10; i++)
	{
		printf("enter your number %d for first array:\n",i+1);
		scanf("%d",a+i);
		printf("enter your number %d for second array:\n",i+1);
		scanf("%d",b+i);
		*(c+i)=*(a+i)+*(b+i);
		if (*(c+i)%2==0)
		{
			(*e)++;
		}
		else
		{
			(*d)++;
		}
	}
	printf("there are %d even number(s) and %d odd number(s) in the final sum.",*e,*d);
	return 0;
}
```

Function:

27. Write a program using function to check whether number is prime or not. (with value, with return).

```
#include<stdio.h>
int prime(int);
int main()
{
	int i,pc;
	printf("enter the number:\n");
	scanf("%d",&i);
	pc=prime(i);
```

```
        if(pc==2)
        {
                printf("prime");
        }
        else if(pc==0)
        {
                printf("zero");
        }
        else if(pc==1)
        {
                printf("one");
        }
        else
        {
                printf("composite");
        }
        return 0;

}
int prime(int num)
{
        int count=0,i;
        for(i=1; i<=num; i++)
        {
                if (num%i==0)
                {
                        count++;

                }
        }
        return count;
}
```

28. Write a program using function to swap two numbers by passing pointers.
(with value, with return).

```
#include<stdio.h>
int swap(int *a,int *b);
int main()
{
        int i,j,*p,*q,pc;
        p=&i;
        q=&j;
        printf("enter the two numbers:\n");
        scanf("%d %d",p,q);
        printf("the correct order is:%d %d\n",*p,*q);
        pc=swap(p,q);
        printf("the swapped order is:%d %d\n",*p,*q);
        return 0;
}
int swap(int *a,int *b)
{
        int temp=*a,*u;
        u=&temp;
        *a=*b;
        *b=*u;
        return 1;
}
```

29. Write a program using function to check whether number is Armstrong or not by passing pointers. (with value, with return)

```
#include<stdio.h>
#include<math.h>
int arm(int *a);
```

```
int main()
{
	int i,*p;
	p=&i;
	printf("enter the the number:\n");
	scanf("%d",p);
	if (arm(p)==0)
	{
		printf("given number %d is not armstrong",*p);
	}
	else if (arm(p)==1)
	{
		printf("given number %d is armstrong",*p);
	}
	return 0;
}
int arm(int *a)
{
	int temp,*t,count=0,*c,k,l,val=0,*v;
	t=&temp;
	c=&count;
	v=&val;
	for (k=*a; k>0; k/=10)
	{
		(*c)++;
	}
	for (k=*a; k>0; k/=10)
	{
		*t=k%10;
		*v=(*v)+pow(*t,*c);
	}
	if(*v==*a)
	{
		return 1;
	}
	else
	{
		return 0;
	}
}
```

30. Write a program to check whether number is Armstrong or not using function. (with value, with return).

```
#include<stdio.h>
#include<math.h>
int arm(int *a);
int main()
{
	int i,*p;
	p=&i;
	printf("enter the the number:\n");
	scanf("%d",p);
	if (arm(p)==0)
	{
		printf("given number %d is not armstrong",*p);
	}
	else if (arm(p)==1)
	{
		printf("given number %d is armstrong",*p);
	}
	return 0;
}
int arm(int *a)
{
	int temp,*t,count=0,*c,k,l,val=0,*v;
```

```
        t=&temp;
        c=&count;
        v=&val;
        for (k=*a; k>0; k/=10)
        {
                (*c)++;
        }
        for (k=*a; k>0; k/=10)
        {
                *t=k%10;
                *v=(*v)+pow(*t,*c);
        }
        if(*v==*a)
        {
                return 1;
        }
        else
        {
                return 0;
        }
}
```

Structure:

Student Record Management

31. Write a C program to manage student records using structure, arrays, and functions. Define a structure named Student that includes roll number, name, marks of 5 subjects (stored in an array), total marks, and percentage. Declare an array to store details of 5 students. Write separate functions to input student details, calculate total and percentage, and display all student information. Finally, write a function to determine and display the topper (student with the highest percentage).

```
#include<stdio.h>
#include<stdlib.h>
struct Student
{
        int Roll_no;
        char Name[50];
        // order of stored marks C of 80,Maths of 80,Technical Communication of 60,FOIT of 80,Sociology of 80
        float Marks[5];
        float Total_Marks;
        float Percentage;
}  Student_Info[5];
void Store();
void total();
void grade();
void display();
void topper();
int i,j;
int main()
{
        Store();
        total();
        grade();
        display();
        topper();
        return 0;
}
void Store()
{
        for(i=0; i<5; i++)
        {
                fflush(stdin);
                printf("enter  name of student %d:", i+1);
                gets(Student_Info[i].Name);
```

```
		printf("enter roll no. of student %s:", Student_Info[i].Name);
		scanf("%d",&Student_Info[i].Roll_no);
		printf("enter marks for student %s in C:", Student_Info[i].Name);
		scanf("%f",&Student_Info[i].Marks[0]);
		printf("enter marks for student %s in Maths:", Student_Info[i].Name);
		scanf("%f",&Student_Info[i].Marks[1]);
		printf("enter marks for student %s in Technical Communication:", Student_Info[i].Name);
		scanf("%f",&Student_Info[i].Marks[2]);
		printf("enter marks for student %s in FOIT:", Student_Info[i].Name);
		scanf("%f",&Student_Info[i].Marks[3]);
		printf("enter marks for student %s in Sociology:", Student_Info[i].Name);
		scanf("%f",&Student_Info[i].Marks[4]);
	}
}
void total()
{
	for(i=0; i<5; i++)
	{
		Student_Info[i].Total_Marks=0;
		for(j=0; j<5; j++)
		{
			Student_Info[i].Total_Marks=Student_Info[i].Total_Marks+Student_Info[i].Marks[j];
		}
	}
}
void grade()
{
	for(i=0; i<5; i++)
	{
		Student_Info[i].Percentage=(Student_Info[i].Total_Marks/380)*100;
	}
}
void display()
{
	printf("\n*************************Student Info Table*****************************\n");
	printf("Name\t\t|Roll No.\t|C\t|Maths\t|Technical Communication\t|FOIT\t|Sociology\t|Total\t|Percentage\n");
	for(i=0; i<5; i++)
	{
		printf("%s\t%d\t",Student_Info[i].Name,Student_Info[i].Roll_no);
		for(j=0; j<5; j++)
		{
			printf("%.1f\t",Student_Info[i].Marks[j]);
		}
		printf("%.1f\t%.2f\t\n",Student_Info[i].Total_Marks,Student_Info[i].Percentage);
	}
}
void topper()
{
	struct Student Highest=Student_Info[0];
	for(i=0; i<5; i++)
	{
		if (Highest.Percentage<Student_Info[i].Percentage)
		Highest=Student_Info[i];
	}
	printf("Congratulations to %s for scoring the highest\n",Highest.Name);
}
```

Employee Salary System

32. Write a C program using structure, arrays, and functions to maintain employee salary records. Define a structure named Employee that includes employee ID, name, basic salary, three types of allowances (stored in an array), and gross salary. Declare an array to store records of 4 employees. Create functions to calculate gross salary, display employees earning more than 50,000, and search for an employee using employee ID.

```
#include<stdio.h>
#include<stdlib.h>
struct Employee
{
        int Employee_ID;
        char Name[50];
        int Basic_Salary;
        int Allowances[3];
        int Gross_Salary;
}  Employee_Record[4];
void total();
void display();
void search();
int i,j;
int main()
{
        for(i=0; i<4; i++)
        {
                fflush(stdin);
                printf("enter  name of Employee %d:", i+1);
                gets(Employee_Record[i].Name);
                printf("enter Employee ID of %s:", Employee_Record[i].Name);
                scanf("%d",&Employee_Record[i].Employee_ID);
                printf("enter Employee Basic Salary of %s:", Employee_Record[i].Name);
                scanf("%d",&Employee_Record[i].Basic_Salary);
                for(j=0; j<3; j++)
                {
                        printf("enter allowance of %s for type %d:",Employee_Record[i].Name,j+1);
                        scanf("%d",&Employee_Record[i].Allowances[j]);

                }
        }
        total();
        display();
        search();
        return 0;
}
void total()
```

```
{
        for(i=0; i<4; i++)
        {
                Employee_Record[i].Gross_Salary=Employee_Record[i].Basic_Salary;
                for(j=0; j<3; j++)
                {
                        Employee_Record[i].Gross_Salary=Employee_Record[i].Gross_Salary+Employee_Record[i].Allowances[j];
                }
        }
}
void display()
{
        printf("\n********************Employee Info Table(Earnings>50,000)********************\n");
        printf("Name\t\t|Employee ID|Basic Salary|Allowance Type 1|Allowance Type 2|Allowance Type 3|Gross Salary\n");
        for(i=0; i<4; i++)
        {
                if(Employee_Record[i].Gross_Salary>50000)
                {
                        printf("%s\t%d\t",Employee_Record[i].Name,Employee_Record[i].Employee_ID);
                        for(j=0; j<3; j++)
                        {
                                printf("%d\t",Employee_Record[i].Allowances[j]);
                        }
                        printf("%d\n",Employee_Record[i].Gross_Salary);
                }
        }
}
void search()
{
        int k=0;
        char response;
        printf("Do you want to search for any Employee?(Y/N)");
        fflush(stdin);
        scanf("%c",&response);
        if (response=='Y'||response=='y')
        {
                int Find_ID;
                printf("enter the ID of the employee you are looking for:");
```

```
                scanf("%d",&Find_ID);
                for(i=0; i<4; i++)
                {
                        if (Find_ID==Employee_Record[i].Employee_ID)
                        {
                                printf("record found >>\t");
                                printf("Name: %s\tEmployee ID:
%d\t",Employee_Record[i].Name,Employee_Record[i].Employee_ID);
                                for(j=0; j<3; j++)
                                {
                                        printf("Allowance Type%d: %d\t",j+1,Employee_Record[i].Allowances[j]);
                                }
                                printf("Gross Salary: %d\n",Employee_Record[i].Gross_Salary);
                                //can be used for same data entered twice or more
                                k++;
                        }
                }
                if(k==0)
                {
                        printf("Data not in the system");
                }
        }
        else
        {
                printf("\nthank you!\n");
        }
}
```

Library Book Management

33. Develop a C program to manage library books using structure, arrays, and functions. Define a structure named Book containing book ID, title, price, and issue status (1 for issued, 0 for available). Store records of 6 books in an array of structures. Write functions to count and display the number of available books, display books with price greater than 500, and issue a book based on its book ID.

```
#include<stdio.h>
#include<stdlib.h>
struct Book
{
        int Book_ID;
        char Title[50];
        int Price;
```

```
        int Issue_Status;
} Book_Record[6];
void available();
void display();
void issue();
int i=0,j=0,count=0;
int main()
{
        for(i=0; i<6; i++)
        {
                fflush(stdin);
                printf("enter  title of Book %d:", i+1);
                gets(Book_Record[i].Title);
                printf("enter Book ID of %s:",Book_Record[i].Title);
                scanf("%d",&Book_Record[i].Book_ID);
                printf("enter price of %s:",Book_Record[i].Title);
                scanf("%d",&Book_Record[i].Price);
                printf("enter issue status for %s:",Book_Record[i].Title);
                scanf("%d",&Book_Record[i].Issue_Status);

        }
        available();
        display();
        issue();
        return 0;
}
void available()
{
        count=0;
        printf("\n*******************Book Info Table(Available)*******************\n");
        printf("Name\t\t|Book ID|Price\n");
        for(i=0; i<6; i++)
        {
                if (Book_Record[i].Issue_Status==0)
                {
                        printf("%s\t%d\t",Book_Record[i].Title,Book_Record[i].Book_ID);
                        printf("%d\n",Book_Record[i].Price);
                        count++;
```

```
                }
        }
        if (count==0)
        {
                printf("no books available for issue\n");
        }
        else
        {
                printf("\nthere are %d book(s) available for issue\n",count);
        }
}
void display()
{
        count=0;
        printf("\n********************Book Info Table(Price>500)********************\n");
        printf("Name\t\t|Book ID|Price\n");
        for(i=0; i<6; i++)
        {
                if (Book_Record[i].Price>500)
                {
                        printf("%s\t%d\t",Book_Record[i].Title,Book_Record[i].Book_ID);
                        printf("%d\n",Book_Record[i].Price);
                        count++;
                }
        }
        if (count==0)
        {
                printf("No books above Rs.500\n");
        }
}
void issue()
{
        int Find_ID;
        printf("enter the ID of the Book you are looking to issue:");
        scanf("%d",&Find_ID);
        for(i=0; i<6; i++)
        {
                if (Find_ID==Book_Record[i].Book_ID)
```

```
                {
                        printf("Book found >>\t");
                        if(Book_Record[i].Issue_Status==0)
                        {
                                printf("Book issue has been confirmed\nBook no longer available");
                                Book_Record[i].Issue_Status=1;
                        }
                        else if (Book_Record[i].Issue_Status==1)
                        {
                                printf("Book is already issued!!\nCan not be issued again!!!");
                        }
                        break;
                }
        }
}
```

Product Inventory System

34. Write a C program to manage product inventory using structure, arrays, and functions. Define a structure named Product that includes product ID, product name, quantity, and price. Declare an array to store 5 products. Write functions to calculate the total inventory value (quantity × price), find the product with the lowest quantity, and update the quantity after selling a certain number of items.

```
#include<stdio.h>
#include<stdlib.h>
struct Product
{
        int Product_ID;
        char Product_Name[50];
        int Quantity;
        int Price;
} Product_Record[5];
void Total_Inventory_Value();
void Lowest_Quantity();
void Update_Inventory();
int i=0,count=0;
int main()
{
        for(i=0; i<5; i++)
        {
                fflush(stdin);
                printf("Enter Product Name:%d:", i+1);
                gets(Product_Record[i].Product_Name);
                printf("Enter Product ID of %s:",Product_Record[i].Product_Name);
                scanf("%d",&Product_Record[i].Product_ID);
                printf("Enter price of %s:",Product_Record[i].Product_Name);
                scanf("%d",&Product_Record[i].Price);
                printf("Enter quantity of %s in stock:",Product_Record[i].Product_Name);
                scanf("%d",&Product_Record[i].Quantity);

        }
        Total_Inventory_Value();
        Lowest_Quantity();
        Update_Inventory();
```

```
	return 0;
}
void Total_Inventory_Value()
{
	count=0;
	for(i=0; i<5; i++)
	{
		count=count+Product_Record[i].Price*Product_Record[i].Quantity;
	}
	printf("Total Inventory Value is:%d",count);
}
void Lowest_Quantity()
{
	struct Product least=Product_Record[0];
	for(i=1; i<5; i++)
	{
		if (Product_Record[i].Quantity<least.Quantity)
		{
			least=Product_Record[i];
		}
	}
	printf("The product with the least inventory is %s, ID:%d with %d quantity valued at
%d",least.Product_Name,least.Product_ID,least.Quantity,least.Price);
}
void Update_Inventory()
{
Flag:
	char check;
	int pass=0;
	count=0;
	int Find_ID;
	printf("enter the ID of the product you are handling:");
	scanf("%d",&Find_ID);
	for(i=0; i<5; i++)
	{
		if (Find_ID==Product_Record[i].Product_ID)
		{
			printf("Product found\nHow many of it were sold?\n");
			scanf("%d",&count);
			if(Product_Record[i].Quantity>=count)
			{
				Product_Record[i].Quantity=Product_Record[i].Quantity-count;
				printf("Sucessfully registered in the system %d stock remain\n",Product_Record[i].Quantity);
			}
			else
			{
				printf("Quantity exceeds inventory is the product ID %d correct\n",Find_ID);
			}
			pass=1;
		}
	}
	if (pass==0)
	{
		printf("Product not found in the system\n");
	}
	fflush(stdin);
	printf("Do you have to update any other inventory?(y/n)\n");
	scanf("%c",&check);
	if(check=='y'||check=='Y')
	{
		goto Flag;
	}
}
```

Cricket Player Statistics

35. Create a C program using structure, arrays, and functions to maintain cricket player statistics. Define a structure named Player that includes player name, number of matches played, runs scored in the last 5 matches (stored in an array), and batting average. Store details of 3 players in an array of structures. Write functions to calculate the batting average, display the player with the highest total runs, and sort the players in descending order based on their average.

```
#include<stdio.h>

#include<stdlib.h>

struct Player

{

	char Player_Name[50];

	int No_of_matches_played;

	int Runs[6];//Index 5 is used to store the total runs made in 5 matches

	int Batting_Average;

} Player_Record[3];

void Calculate_Average()

{

	count=0;

	for(i=0; i<3; i++)

	{

		for (j=0; j<5; j++)

		{

			count=count+Player_Record[i].Runs[j];

		}

		Player_Record[i].Runs[5]=count;

		Player_Record[i].Batting_Average=count/5;

		count=0;

	}

}

void Display_Highest()

{

	struct Player highest=Player_Record[0];

	for (i=1; i<3; i++)

	{

		if (highest.Runs[5]<Player_Record[i].Runs[5])

		{

			highest=Player_Record[i];

		}

	}

	printf("The highest no. of runs is %d made my %s.\n",highest.Runs[5],highest.Player_Name);
```

```
}
void Player_Sort()
{
	struct Player temp;
	for (i=0; i<3; i++)
	{
		for (j=i+1; j<3; j++)
		{
			if (Player_Record[i].Batting_Average<Player_Record[j].Batting_Average)
			{
				temp=Player_Record[i];
				Player_Record[i]=Player_Record[j];
				Player_Record[j]=temp;
			}
		}
	}
	printf("\n**********************************PLayer Info Table***************************************\n");
	printf("Player Name\t\t|No. of matches played|Run I|Runs II|Runs III|Runs IV|Runs V|Batting average|\n");
	for (i=0; i<3; i++)
	{
		printf("%s\t%d\t\t",Player_Record[i].Player_Name,Player_Record[i].No_of_matches_played);
		for(j=0; j<5; j++)
		{
			printf("%d\t",Player_Record[i].Runs[j]);
		}
		printf("%d\n",Player_Record[i].Batting_Average);
	}
}
int i=0,j=0,count=0;
int main()
{
	for(i=0; i<3; i++)
	{
		fflush(stdin);
		printf("Enter Player Name:%d:", i+1);
		gets(Player_Record[i].Player_Name);
		printf("Enter No. of games played by %s:",Player_Record[i].Player_Name);
		scanf("%d",&Player_Record[i].No_of_matches_played);
```

```
		for (j=0; j<5; j++)
		{
			printf("Enter no. of runs made by %s on game %d:",Player_Record[i].Player_Name,j+1);
			scanf("%d",&Player_Record[i].Runs[j]);
		}
	}
	Calculate_Average();
	Display_Highest();
	Player_Sort();
	return 0;
}
```